

Ajitkumar Senthil Kumar

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Summary

Full Stack engineer with 2.4 years of experience building innovative web applications and RESTful APIs. Proficient in Python, React, and FastAPI, with a strong background in developing production-ready solutions and automating complex workflows. Demonstrates a proven ability to enhance system performance and user experience through effective full-stack development.

Education

State University of New York at Buffalo | *M.S., Data Science (GPA: 3.3/4)* **2024 - 2025**

Sri Sairam Engineering College, Tambaram | *B.E., Mechanical Engineering (GPA: 8.1/10)* **2017 - 2021**

Experience

Human CloudSoft | *Software Intern* **May 2023 - Jun 2024**

- Developed a web-based inventory platform that automated stock tracking, transitioning all stock logs from manual to digital.
- Enhanced inventory accuracy by 35% and reduced audit time by 40% by implementing automated alerts and streamlining workflows.

Klicknet Info Services Pvt Ltd | *Frontend Developer* **Apr 2022 - Apr 2023**

- Engineered responsive and accessible UIs using React and TypeScript, achieving a 25% increase in user retention.
- Collaborated with backend teams to optimize workflows and performance, boosting cross-team efficiency by 30%.

L&T Rubber Processing Machinery | *Design Engineer* **Nov 2021 - Feb 2022**

- By converting complex 3D models into precise 2D AutoCAD drawings, I ensured production readiness and helped maintain defect-free outputs.
- Reduced manufacturing errors by 15% and rework by 20% through detailed component modeling in CREO and the application of GD&T principles.

Projects

SmartLoanPred **Sep 2024 - Present**

- Developed a production-ready full-stack application for real-time loan approval prediction, using a React + Tailwind frontend and a FastAPI backend with secure JWT/OAuth authentication.
- Deployed the Dockerized services via a GitHub Actions CI/CD pipeline with an Nginx reverse proxy.

MIDAS **Jul 2024 - Aug 2024**

- Developed an end-to-end pipeline in PyTorch for dermatology image classification and risk screening, applying extensive image preprocessing, data augmentation with stratified sampling, and class-weighted loss to improve performance on a diverse dataset.
- Architecting the transition from a research model to an ongoing product by implementing MLflow for scalable experiment tracking and model lifecycle management, while integrating Grad-CAM as a core feature to ensure interpretability and trust for medical professionals.

CovDet **May 2024 - Jun 2024**

- Built a computer vision classifier to detect respiratory diseases from X-ray images, performing extensive image cleaning and augmentation while utilizing early stopping and cosine annealing to train robust, high-performance models.
- Engineered a production-ready inference pipeline by exporting final models to efficient TorchScript/ONNX formats and deploying them via a FastAPI endpoint for real-time analysis.

New York Taxi Rides **Feb 2024 - Apr 2024**

- Modeled NYC taxi ride demand using historical trip data through extensive feature engineering with lag variables, rolling windows, and external joins (holidays, weather).
- Used time-based cross-validation to ensure model robustness and integrated results into a Hopworks feature store.
- Developed a Streamlit dashboard for real-time monitoring and automated retraining workflows using GitHub Actions.

Technical Skills

- **Programming:** Python, R, SQL, JavaScript, TypeScript
- **ML/DS:** scikit-learn, PyTorch, Pandas, NumPy, imbalanced-learn, MLflow, Jupyter
- **Frontend:** React, Next.js, Tailwind CSS, shadcn/ui
- **Backend:** FastAPI, SQLAlchemy, Auth (JWT/OAuth)
- **Databases:** PostgreSQL, SQL Server, Supabase, SQLite
- **DevOps/Cloud:** Docker, GitHub Actions, Nginx, AWS, DigitalOcean
- **RAG/Agents:** LangChain, FAISS/Chroma, Ollama